ModelIt!

Student Activity Pack

Engineering a Superbug Killer

Optimizing Bacteriophage Therapy Against Antibiotic-Resistant Bacteria

Grade 09L3 | NGSS HS-LS1-2, HS-LS4-2 | Cause and Effect / Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why are antibiotic-resistant bacteria considered one of the greatest threats to human health in the 21st century?**

2. **What is a bacteriophage and how is it different from the viruses that make humans sick?**

3. **Draw a diagram showing how you think a virus could be used as a medical treatment instead of a disease.**

4. **What do you think happens when bacteria evolve resistance to a treatment — does the bacteria 'decide' to become resistant?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Phage Population
- Target Bacteria Population
- Phage Specificity
- Bacterial Resistance Rate
- Immune System Response
- Phage Replication Rate
- Healthy Microbiome Impact
- Treatment Timing
- Dosage Optimization

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Standard Protocol

Moderate dose, medium specificity, early treatment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Delayed Treatment

Late timing, established bacterial resistance

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Optimized Cocktail

High specificity, high dose, early timing

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Phage-Antibiotic Synergy — The phenomenon where bacteria that evolve resistance to phages become re-sensitized to antibiotics, and vice versa — creating a therapeutic one-two punch

Biofilm Penetration — The ability of phages to penetrate bacterial biofilms — dense protective communities that antibiotics cannot reach — using phage-encoded depolymerase enzymes

Phage Engineering Level — The degree of genetic modification applied to the phage genome — from wild-type phages to CRISPR-enhanced phages with expanded host range and resistance-countering capabilities

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Antibiotics are failing. Can we engineer viruses to kill the superbugs that antibiotics can't? Use evidence from your simulation to support your answers.

1 How does your model demonstrate the predator-prey dynamics between phages and bacteria?

2 Why is phage therapy described as 'self-amplifying' and how does this differ from antibiotic treatment?

3 What trade-offs did you discover between Phage Specificity and treatment speed?

4 How does natural selection drive bacterial resistance to phages, and why might this resistance come with a fitness cost?

5 What are the limitations of modeling phage therapy with nine components when real infections involve thousands of bacterial strains and immune factors?

8 STEM Challenge: Design a Personalized Phage Therapy Protocol

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YOUR ENGINEERING MISSION

Design a treatment protocol that matches engineered phage cocktails to specific patient infections, optimizing timing, dosage, and specificity for maximum effectiveness.

THE SCENARIO

A biotech startup is developing a phage therapy clinic for patients with antibiotic-resistant infections. Each patient's infection is different — different bacterial strains, different resistance profiles, different immune status. Your team must design the decision-making protocol that determines which phages to use, when to start treatment, and how to adjust the approach if resistance emerges.

Engineering Design Thinking

1. WOULD YOU USE A SINGLE PHAGE OR A COCKTAIL OF MULTIPLE PHAGES — AND WHAT ARE THE TRADE-OFFS?
2. HOW WOULD YOU SEQUENCE PHAGE THERAPY WITH EXISTING ANTIBIOTIC TREATMENTS?
3. WHAT BACKUP STRATEGIES WOULD YOU INCLUDE IF THE FIRST PHAGE COCKTAIL ENCOUNTERS RESISTANCE?
4. HOW WOULD YOU ENSURE PHAGE SPECIFICITY IS HIGH ENOUGH TO PROTECT THE PATIENT'S MICROBIOME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Phage Biologists and Phage Therapy Researchers develop bacteriophage treatments for antibiotic-resistant infections. They work for biotech startups, academic medical centers, and government agencies like DARPA, earning $75,000–$150,000/year. Synthetic Biologists who engineer phage genomes for therapeutic applications earn $100,000–$200,000/year.